UNIVERSITÀ DEGLI STUDI DI MILANO

DIPARTIMENTO DI CHIMICA

Semiclassical Initial Value Representation: From large molecular systems to nonadiabatic dynamics

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Dipartimento di Chimica,
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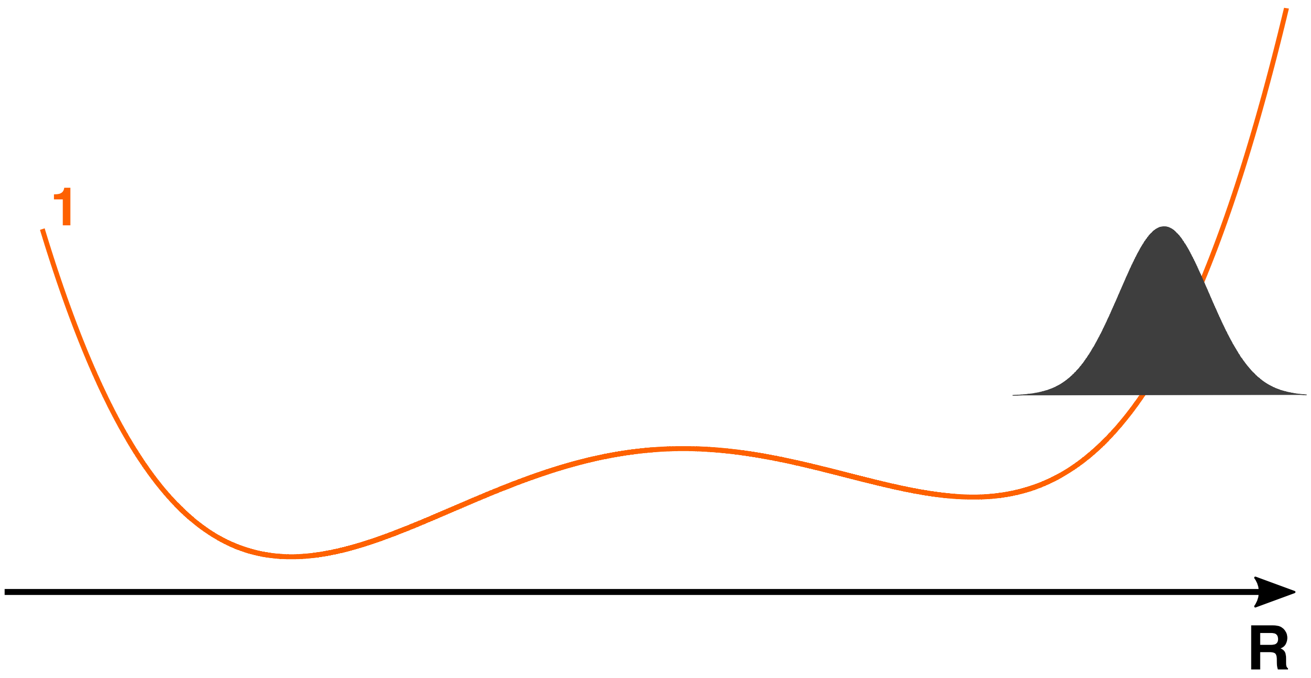
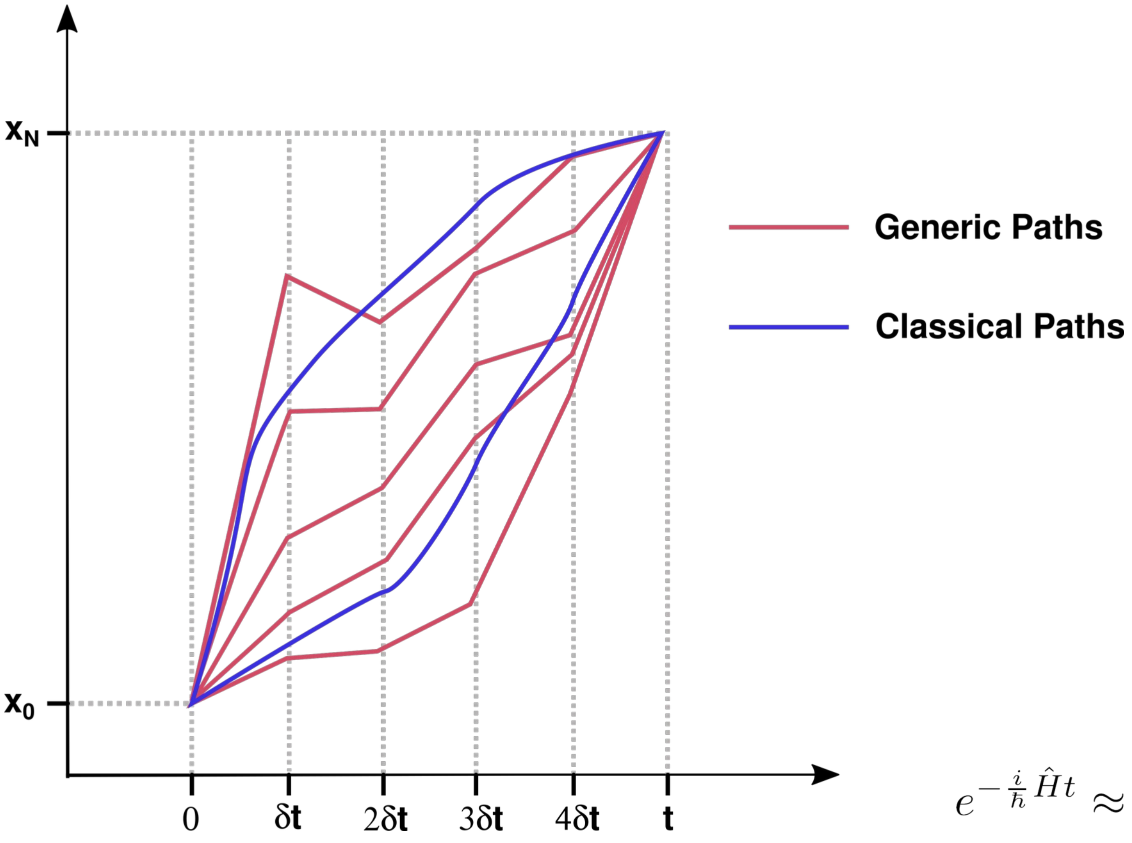
CEOTTO GROUP

Adiabatic dynamics



Semiclassical Molecular Dynamics



$$\langle q_t | e^{-\frac{i}{\hbar}\hat{H}t} | q_0 \rangle = \int \mathcal{D}[q(t)] e^{\frac{i}{\hbar}S(q_0,q_t)}$$

$$\langle q_t | e^{-\frac{i}{\hbar}\hat{H}t} | q_0 \rangle \approx \sum_{clps} \int \mathcal{D}[q(t)] e^{\frac{i}{\hbar}\left[S_t^{cl}+\delta S_t^{cl}+\delta^2 S_t^{cl}\right]}$$

$$e^{-\frac{i}{\hbar}\hat{H}t} \approx \iint \frac{d\mathbf{p}_0 d\mathbf{q}_0}{(2\pi\hbar)^F} C_t(\mathbf{p}_0,\mathbf{q}_0) | \mathbf{p}_t, \mathbf{q}_t \rangle \langle \mathbf{p}_0, \mathbf{q}_0 | e^{\frac{i}{\hbar}S_t(\mathbf{p}_0,\mathbf{q}_0)}$$

J. H. Van Vleck, PNAS, vol. 14, p. 178–188, Feb. 1928.
W. H. Miller, JCP, vol. 53, p. 3578–3587, Nov. 1970.
E. J. Heller, JCP, vol. 75, p. 2923–2931, Sept. 1981.
M. F. Herman and E. Kluk, Chemical Physics, vol. 91, p. 27–34, Nov. 1984.
K. G. Kay, JCP, vol. 101, p. 2250–2260, Aug. 1994.

2942 *J. Phys. Chem. A* **2001,** *105,* 2942−2955

FEATURE ARTICLE

The Semiclassical Initial Value Representation: A Potentially Practical Way for Adding Quantum Effects to Classical Molecular Dynamics Simulations

William H. Miller

Department of Chemistry and Kenneth S. Pitzer Center for Theoretical Chemistry, University of California, and Chemical Sciences Division, Lawrence Berkeley National Laboratory, Berkeley, California 94720-1460

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Some issues:

- Number of trajectories
- Curse of dimensionality
- Hessian matrices

Semiclassical vibrational spectroscopy

$$I(E) = \frac{\Re}{2\pi\hbar} \int_{\mathbb{R}} dt \; e^{\frac{i}{\hbar}Et} \left\langle \Psi(0) \middle| e^{-\frac{i}{\hbar}\hat{H}t} \middle| \Psi(0) \right\rangle$$

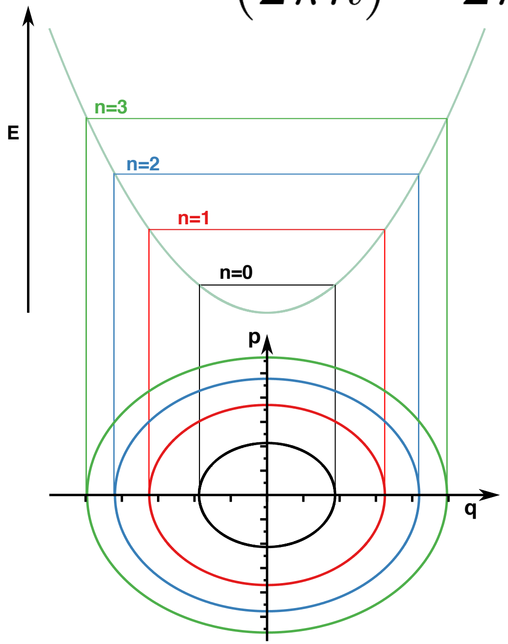
$$\left\langle A \right\rangle_{ps} = \iint d\mathbf{p}_0 d\mathbf{q}_0 \; A(\mathbf{p}_0, \mathbf{q}_0) = \iint d\mathbf{p}_0 d\mathbf{q}_0 \; \frac{1}{\mathrm{T}} \int_0^{\mathrm{T}} dt \; A(\mathbf{p}_t, \mathbf{q}_t) = \left\langle A \right\rangle_{TA}$$

$$I(E) \approx \iint \frac{d\mathbf{p}_0 d\mathbf{q}_0}{(2\pi\hbar)^F} \frac{1}{2\pi\hbar T} \left| \int_0^T dt \; e^{\frac{i}{\hbar}[S_t(\mathbf{p}_0,\mathbf{q}_0)+\phi_t(\mathbf{p}_0,\mathbf{q}_0)+Et]} \left\langle \mathbf{p}_{eq}, \mathbf{q}_{eq} \middle| \mathbf{p}_t, \mathbf{q}_t \right\rangle \right|^2$$

A. L. Kaledin and W. H. Miller, JCP, vol. 118, pp. 7174–7182, Apr. 2003
A. L. Kaledin and W. H. Miller, JCP, vol. 119, pp. 3078–3084, Aug. 2003
Conte, R. and Ceotto, M. (2020). Semiclassical Molecular Dynamics for Spectroscopic Calculations. In Quantum Chemistry and Dynamics of Excited States (eds L. González and R. Lindh)

Divide and Conquer Semiclassical vibrational spectroscopy

$$\tilde{I}(E) = \frac{1}{(2\pi\hbar)^M}\frac{1}{2\pi\hbar T}\left|\int_0^T dt\; e^{\frac{i}{\hbar}\left[\tilde{S}_t(\tilde{\mathbf{p}}_0,\tilde{\mathbf{q}}_0)+\tilde{\phi}_t(\tilde{\mathbf{p}}_0,\tilde{\mathbf{q}}_0)+Et\right]}\left\langle\Psi\middle|\tilde{\mathbf{p}}_t,\tilde{\mathbf{q}}_t\right\rangle\right|^2$$

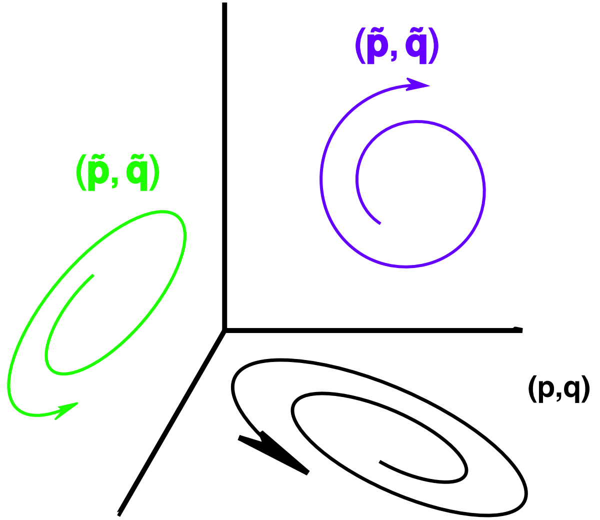


$$\begin{cases} p_J(0) &= \sqrt{\hbar\omega_J(2n_J+1)}\,\sin(\theta_J) \\ q_J(0) &= q_{J,eq}+\sqrt{\frac{\hbar}{\omega_J}(2n_J+1)}\,\cos(\theta_J) \end{cases}$$

N. De Leon and E. J. Heller, JCP, 78, 4005–4017, Mar 1983.
M. Ceotto, D. Dell'Angelo, and G. F. Tantardini; JCP, vol. 133, p. 054701, Aug. 2010.
R. Conte, A. Aspuru-Guzik, and M. Ceotto; JPCL, vol. 4, pp. 3407–3412, Oct. 2013.
M. Ceotto, G. Di Liberto, and R. Conte; PRL, vol. 119, July 2017.
G. Di Liberto, R. Conte, and M. Ceotto; JCP, vol. 148, p. 014307, Jan. 2018.

Divide and Conquer Semiclassical vibrational spectroscopy

$$\tilde{I}(E) = \frac{1}{(2\pi\hbar)^M}\frac{1}{2\pi\hbar T}\left|\int_0^T dt\; e^{\frac{i}{\hbar}\left[\tilde{S}_t(\tilde{\mathbf{p}}_0,\tilde{\mathbf{q}}_0)+\tilde{\phi}_t(\tilde{\mathbf{p}}_0,\tilde{\mathbf{q}}_0)+Et\right]}\left\langle\Psi\middle|\tilde{\mathbf{p}}_t,\tilde{\mathbf{q}}_t\right\rangle\right|^2$$



$$\left|\Psi\right\rangle = \prod_{j=1}^{M}\left[\varepsilon_{1,j}\left|p_{j,eq},q_{j,eq}\right\rangle + \varepsilon_{2,j}\left|-p_{j,eq},q_{j,eq}\right\rangle\right]$$

Enhance the signal of specific normal modes

M. Ceotto, D. Dell'Angelo, and G. F. Tantardini; JCP, vol. 133, p. 054701, Aug. 2010.
R. Conte, A. Aspuru-Guzik, and M. Ceotto; JPCL, vol. 4, pp. 3407–3412, Oct. 2013.
M. Ceotto, G. Di Liberto, and R. Conte; PRL, vol. 119, July 2017.
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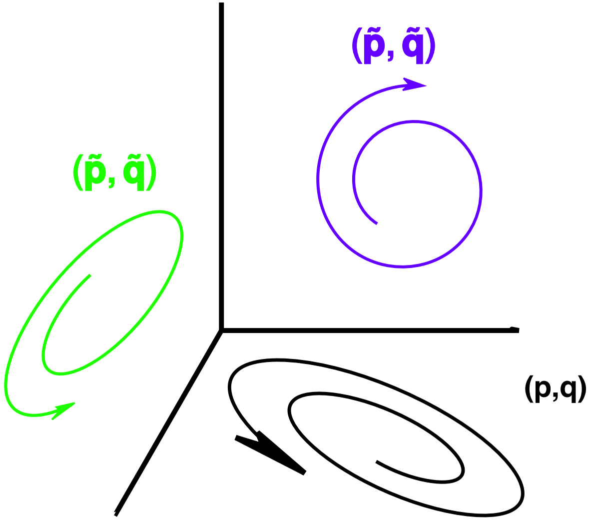
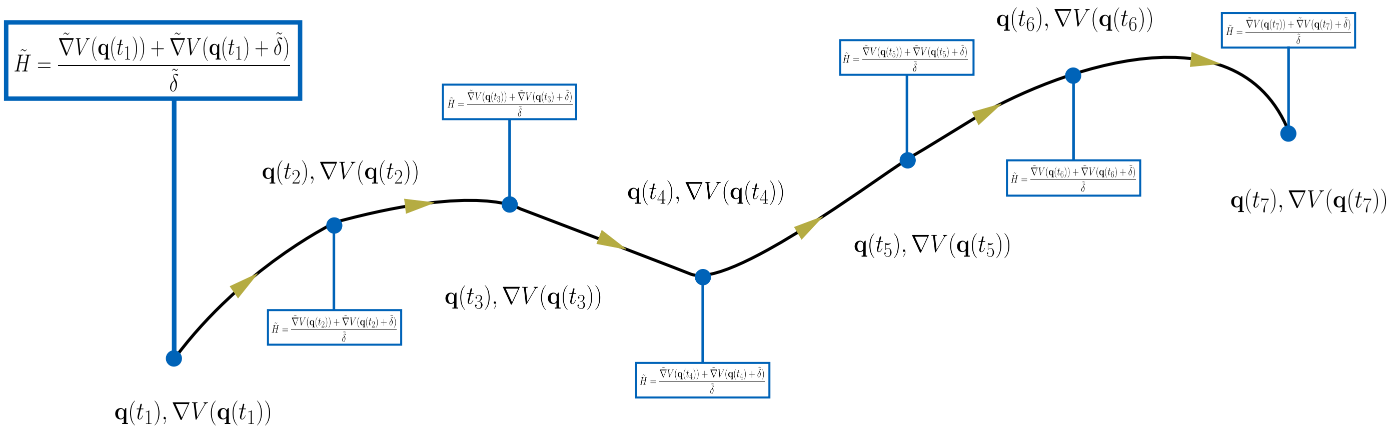
Divide and Conquer Semiclassical vibrational spectroscopy

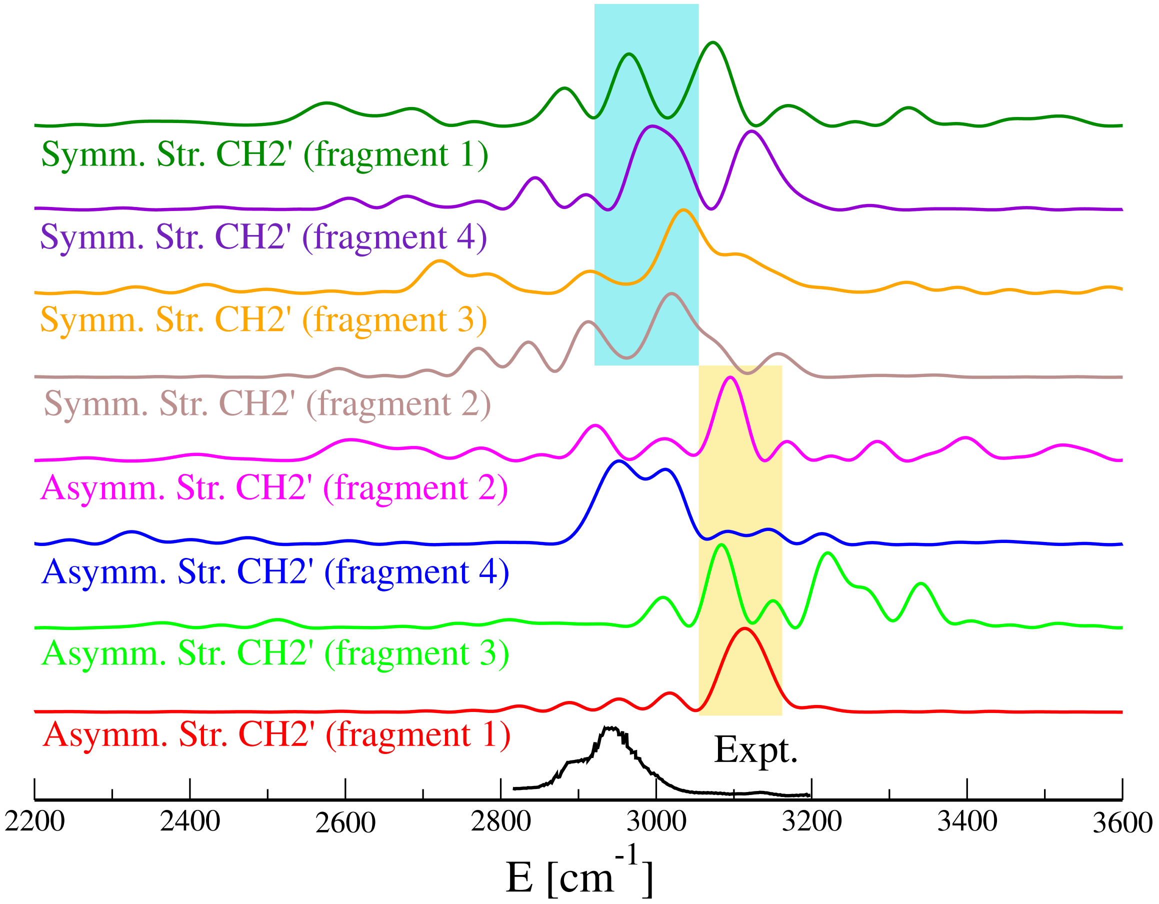
$$\tilde{I}(E) = \frac{1}{(2\pi\hbar)^M} \frac{1}{2\pi\hbar T} \left| \int_0^T dt \; e^{\frac{i}{\hbar}\left[\tilde{S}_t(\tilde{\mathbf{p}}_0,\tilde{\mathbf{q}}_0)+\tilde{\phi}_t(\tilde{\mathbf{p}}_0,\tilde{\mathbf{q}}_0)+Et\right]} \left\langle \Psi \middle| \tilde{\mathbf{p}}_t, \tilde{\mathbf{q}}_t \right\rangle \right|^2$$

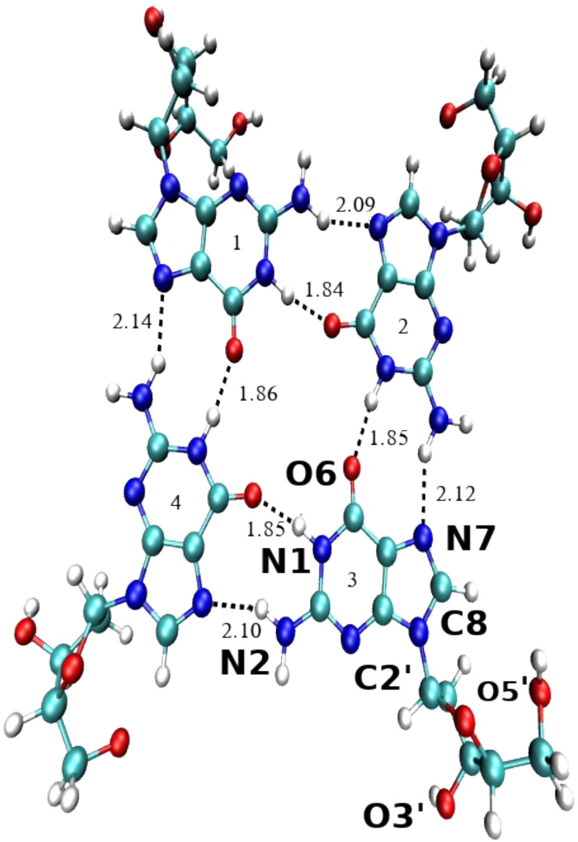




R. Conte, F. Gabas, G. Botti, Y. Zhuang, M. Ceotto; J. Chem. Phys. 150, 244118 (2019)
M. Cazzaniga, M. Micciarelli, F. Moriggi, A. Mahmoud, F. Gabas, M. Ceotto; J. Chem. Phys. 152, 104104 (2020)
M. Gandolfi, M. Ceotto; J. Chem. Theory Comput. 2021, 17, 11, 6733–6746

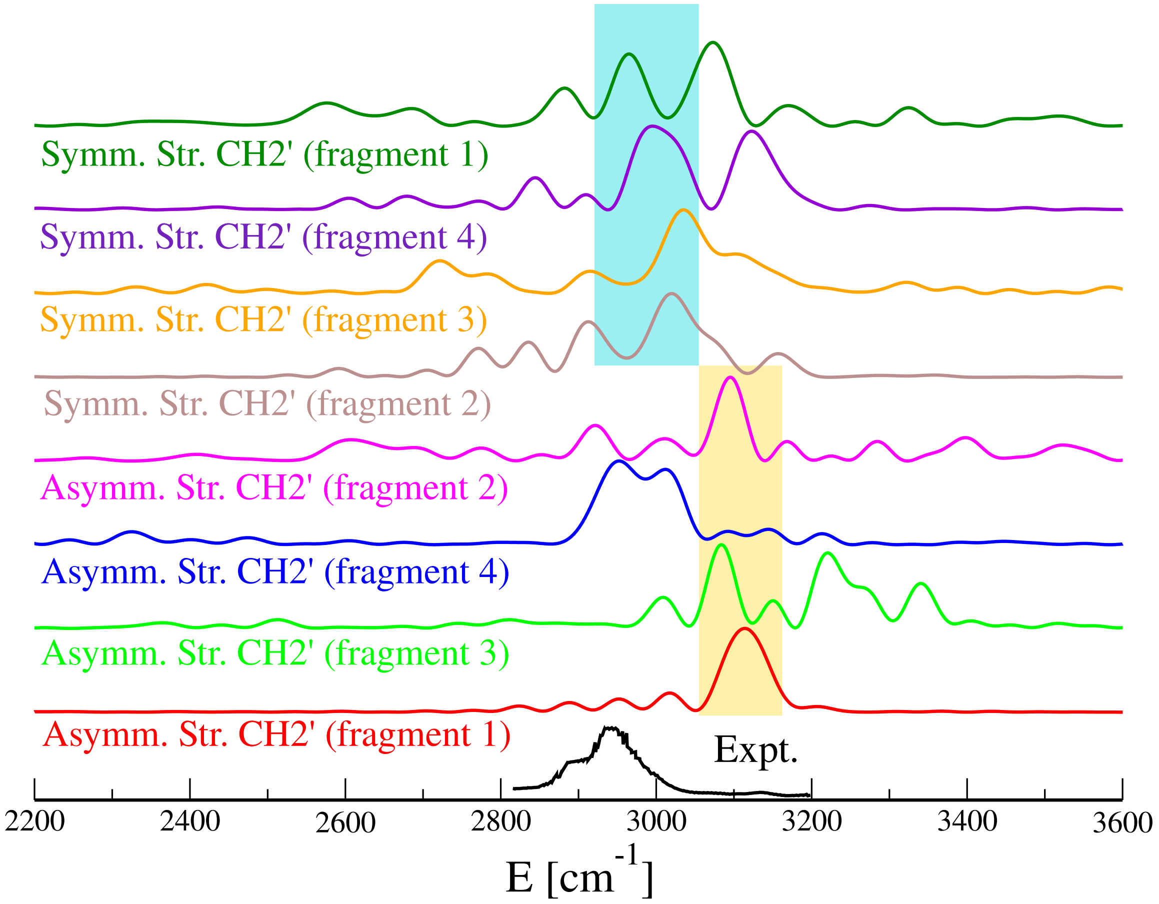
Semiclassical vibrational spectroscopy in condensed phase

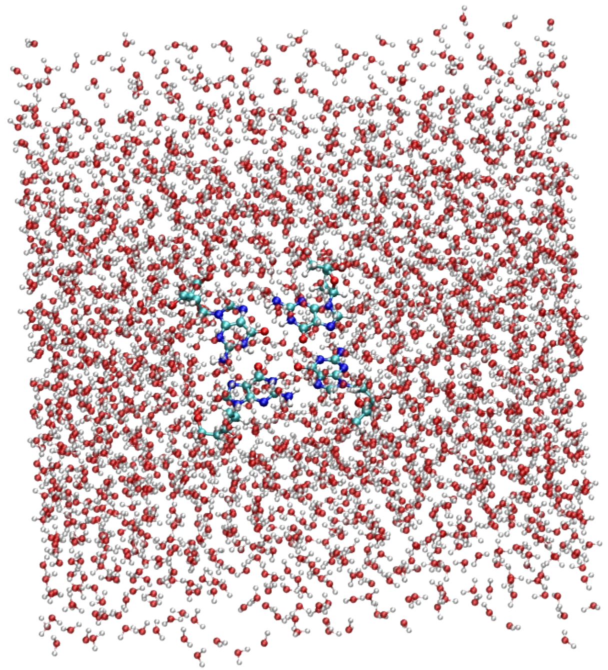




***Moscato D.**; Gabas F. ; Conte R.; Ceotto M.; Journal of Biomolecular Structure and Dynamics 41(23), 1428-14258*
Exp. Spectrum: CV Pagba, SM Lane, S Wachsmann-Hoglu, J. Raman Spectrosc. 2010, 41, 241–247

Semiclassical vibrational spectroscopy in condensed phase





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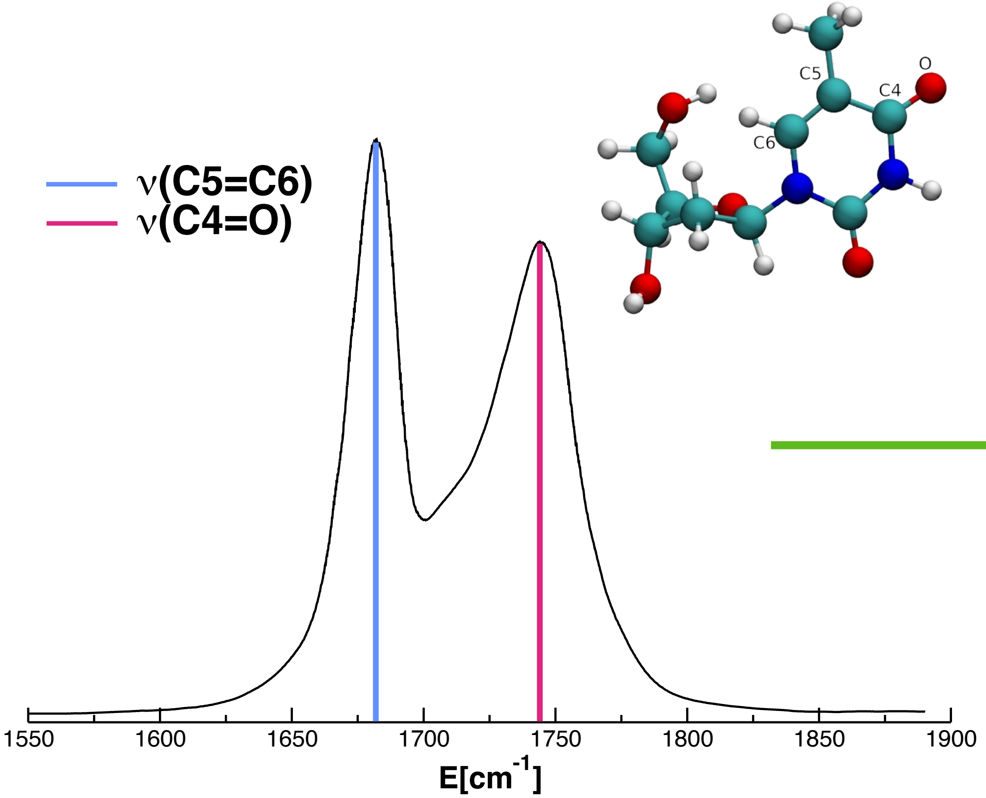
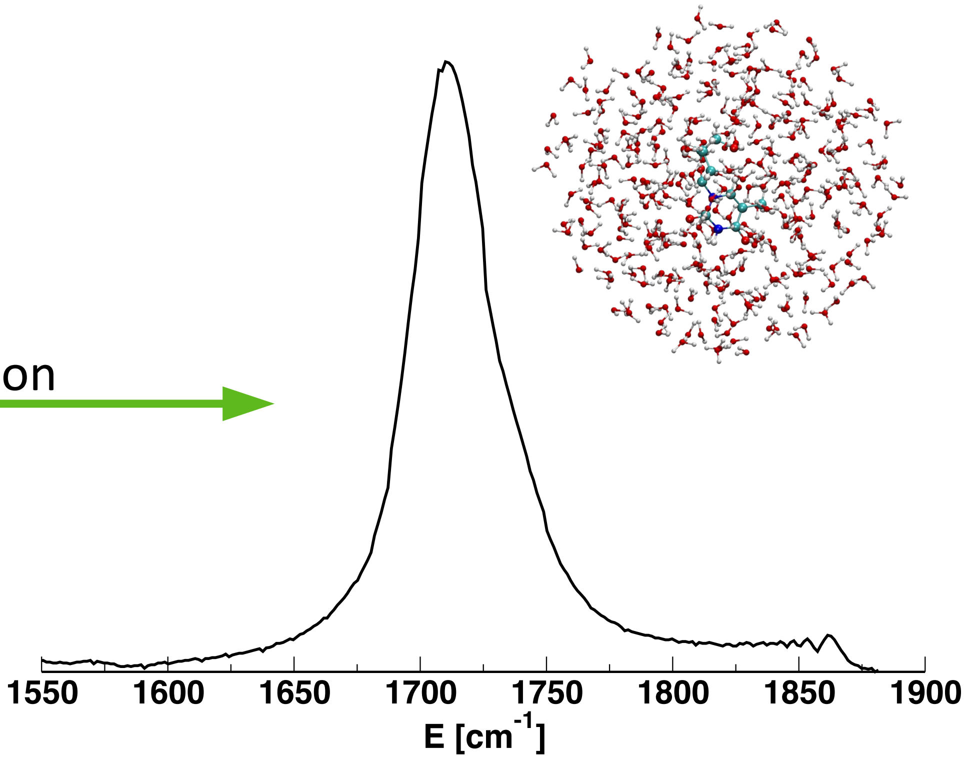


Solvation



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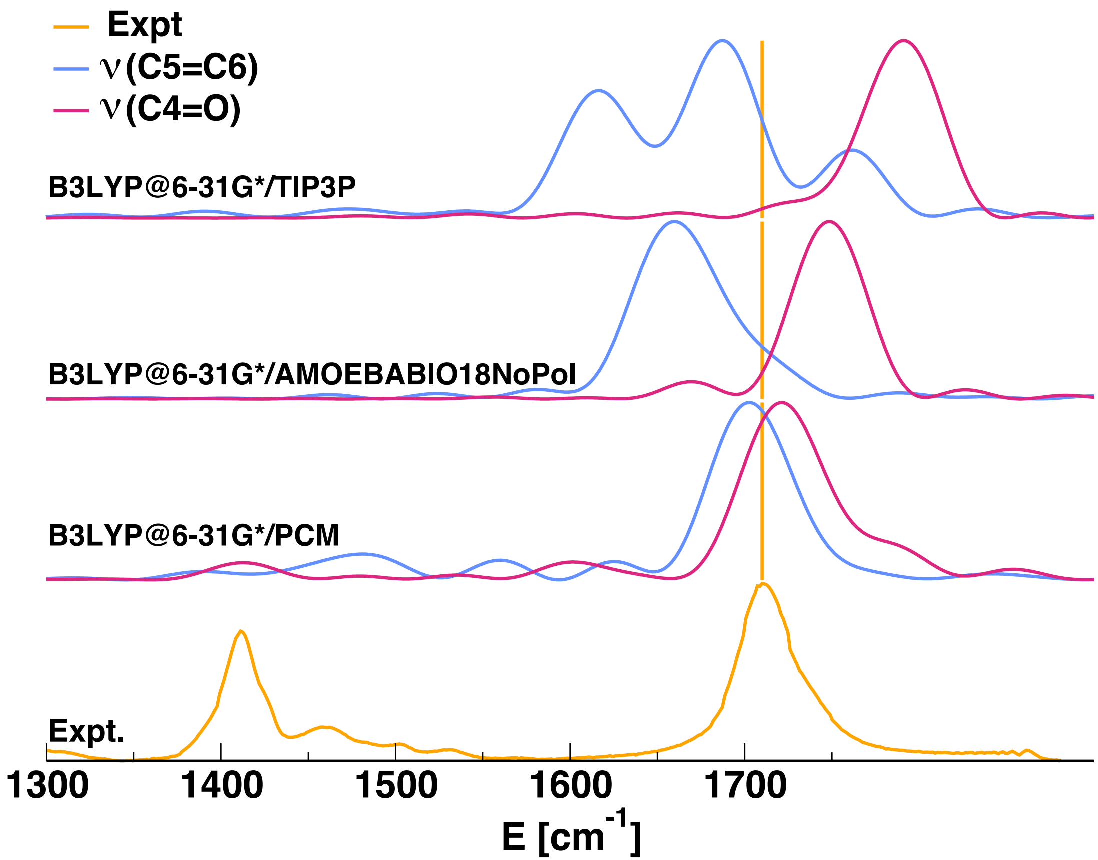
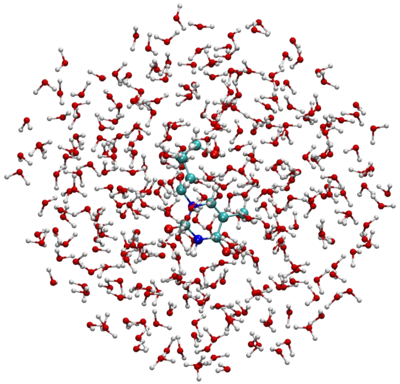
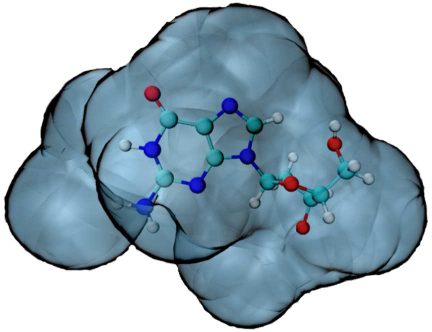
***Moscato D.**; Mandelli G. ; Bondanza M.; Lipparini F.; Conte R.; Mennucci B.; Ceotto M.; JACS, 146, 12, 8179–8188 (2024)*
Exp. Spectrum: L. Beyere; P. Arboleda; V. Monga; G. Loppnow; Can. J. Chem. 2004, 82, 1092-1101

Semiclassical vibrational spectroscopy in condensed phase







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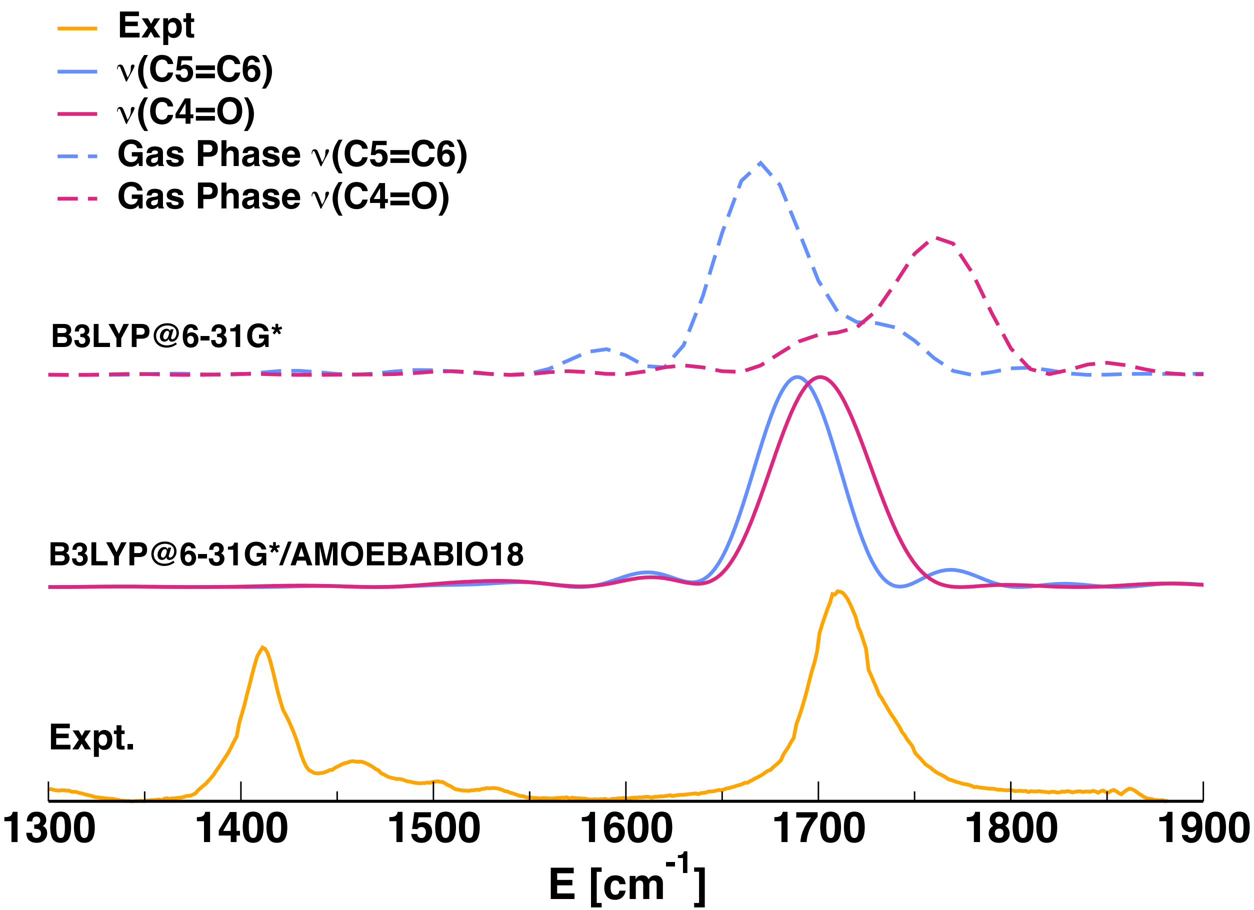
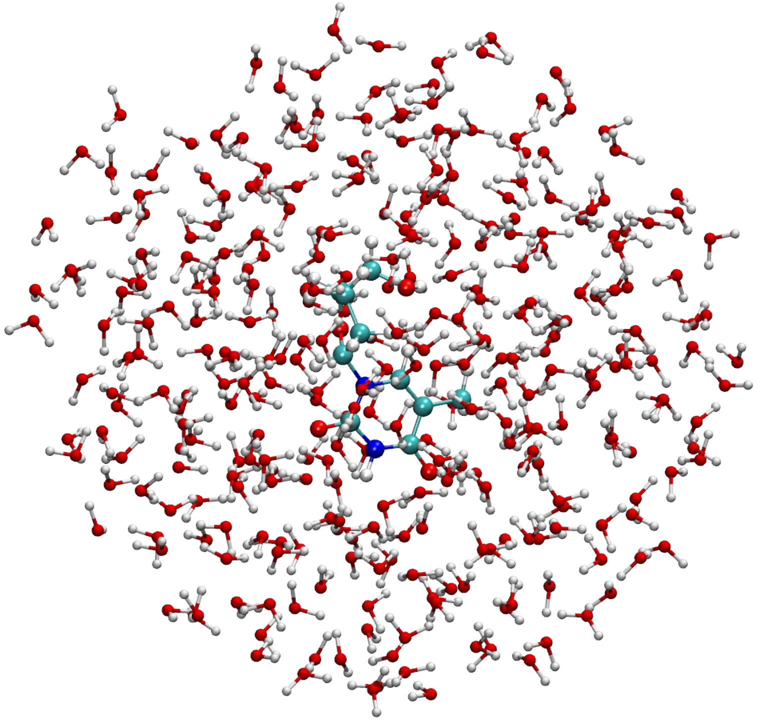
Semiclassical polarizable QM/MM





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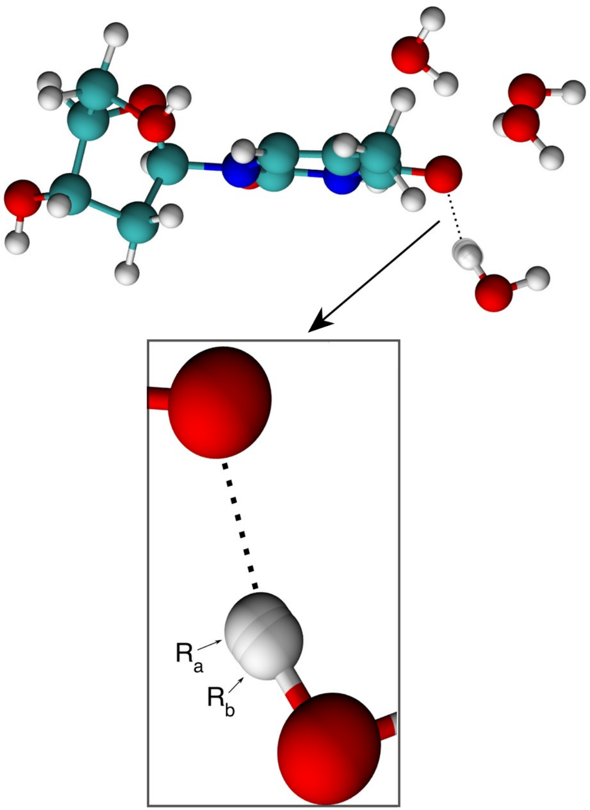
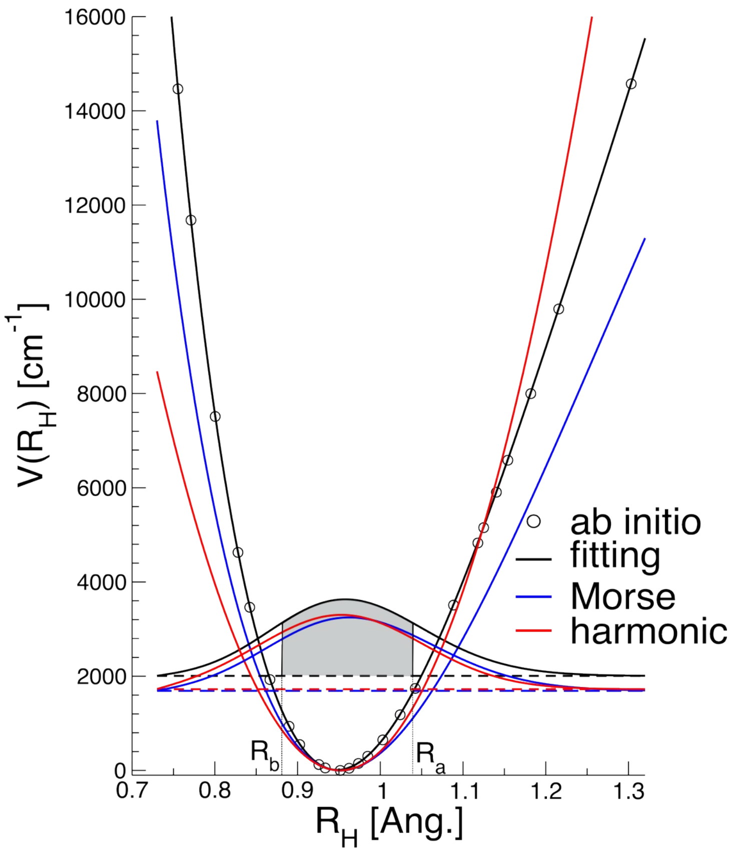
Semiclassical polarizable QM/MM

GEOMETRIES	E_{exch} / [kJ/mol]*	%
Equilibrium	7602.40	0
Stretched	7759.64	+2.07
R_a	7893.09	+3.68
R_b	7591.85	-0.13





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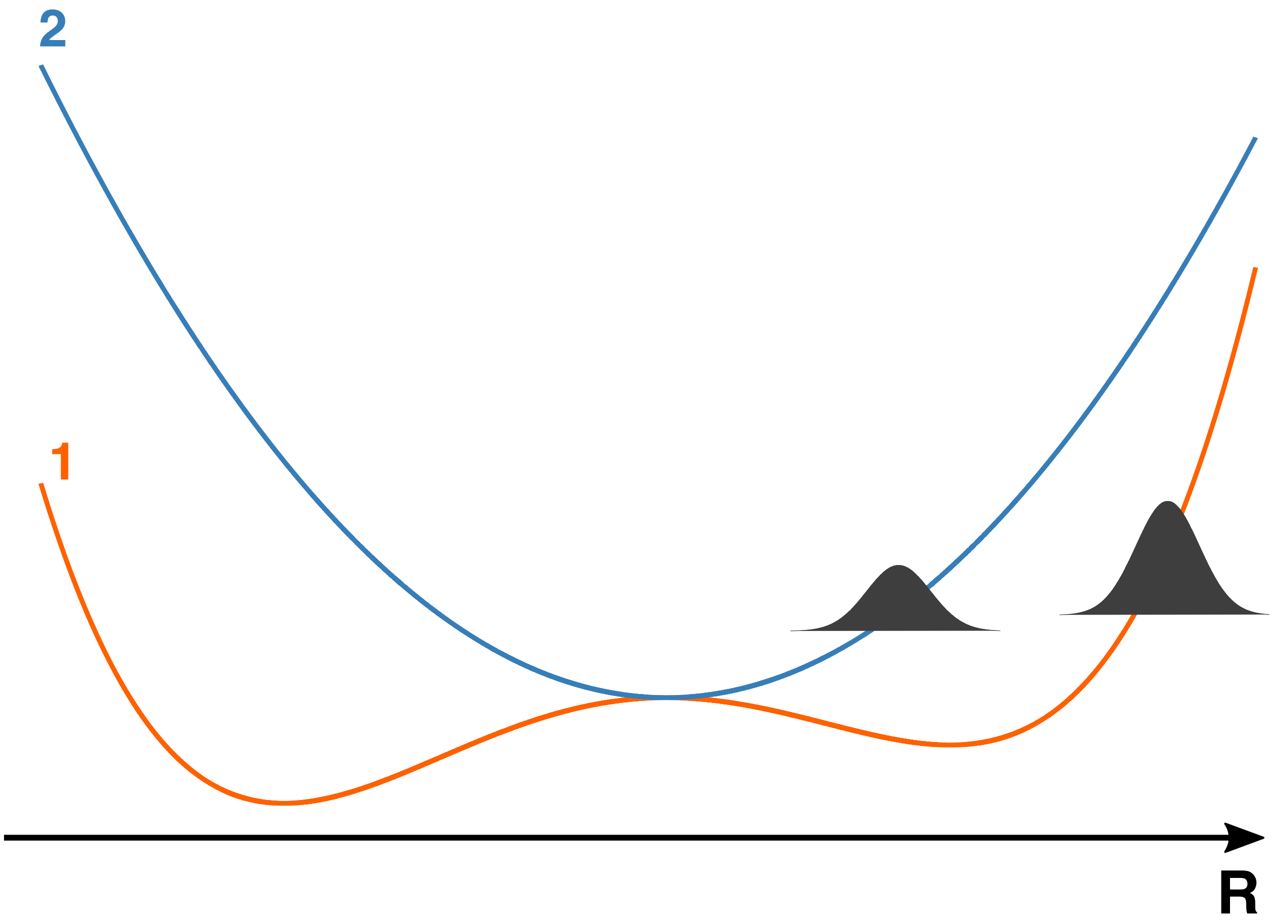
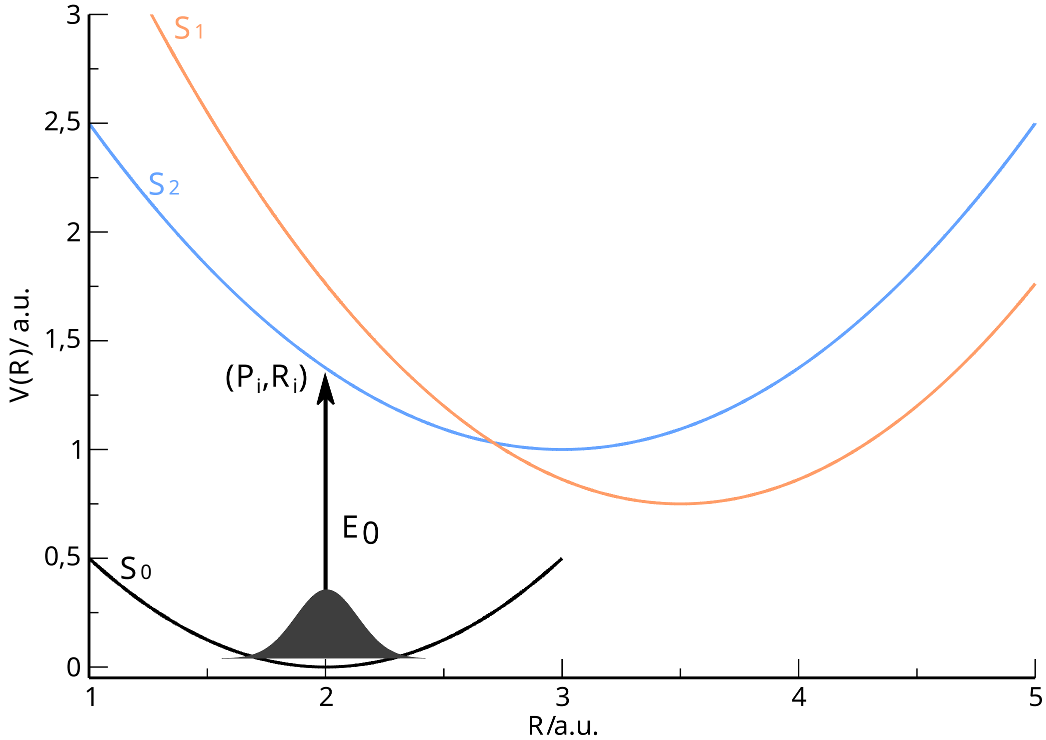
Nonadiabatic semiclassical dynamics



Nonadiabatic semiclassical spectroscopy



$$\sigma_\beta(E) = \frac{E}{2\hbar^2 c\varepsilon_0}\Re\int_{\mathbb{R}} dt\ \mathrm{Tr}\left[\frac{e^{-\beta\hat{H}_0}}{Z(\beta)} e^{\frac{i}{\hbar}\hat{H}_0 t}\hat{\mu} e^{-\frac{i}{\hbar}\hat{\mathbf{H}}_d t}\hat{\mu}\right] e^{\frac{i}{\hbar}Et}$$

$$\lim_{\beta\to+\infty}\frac{e^{-\beta\hat{H}_0}}{Z(\beta)} = |\psi_0\rangle\langle\psi_0|$$

$$\sigma(E) = \frac{E}{2\hbar^2 c\varepsilon_0}\Re\int_{\mathbb{R}} dt\ \langle\psi_0|\hat{\mu} e^{-\frac{i}{\hbar}\hat{\mathbf{H}}_d t}\hat{\mu}|\psi_0\rangle e^{\frac{i}{\hbar}(E+E_0)t}$$

$$\hat{\mu}(\mathbf{R}) \approx \hat{\mu}(\mathbf{R}_{eq}^{(0)})$$

$$\sigma(E) = \underbrace{\frac{E\,|\hat{\mu}|^2}{2\hbar^2 c\varepsilon_0}}_{\text{Lineshape}} \underbrace{\Re\int_{\mathbb{R}} dt\ \langle\psi_0|e^{-\frac{i}{\hbar}\hat{\mathbf{H}}_d t}|\psi_0\rangle e^{\frac{i}{\hbar}(E+E_0)t}}_{\text{Density of states}}$$

J. O. Richardson, P. Meyer, M. O. Pleinert, M. Thoss, Chemical Physics, 482, 124-134, 2017
T. Begusic and J. Vanicek, JPCL, 12, 11, 2997–3005, 2021
Z. T. Zhang J. Vanicek, JCP, 160, 8, 2024.
C. Lanzi, C. Aieta, M. Ceotto, and R. Conte, JCP, 160, 214107, June 2024.
***D. Moscato**, M. Gandolfi and M. Ceotto; manuscript submitted*

Nonadiabatic semiclassical dynamics

$$\hat{H}_{el} = \sum_{l,m}^{N} |l\rangle V_{lm}(\mathbf{R}) \langle m|$$

$$|l\rangle\langle m| \mapsto \hat{a}_l^\dagger \hat{a}_m$$
$$|l\rangle \mapsto |0_1 \ldots 1_l \ldots 0_N\rangle$$

$$\hat{H}_{el} = \sum_{l,m}^{N} \frac{1}{2}(\hat{x}_l - i\hat{p}_l) V_{lm}(\mathbf{R}) (\hat{x}_m + i\hat{p}_m)$$

$$H(\mathbf{P}, \mathbf{R}, \mathbf{p}, \mathbf{x}) = \sum_{J=1}^{F} \frac{P_J^2}{2m_J} + V_0(\mathbf{R}) + \frac{1}{2}\mathbf{x}^T \mathbf{V}(\mathbf{R})\mathbf{x} + \frac{1}{2}\mathbf{p}^T \mathbf{V}(\mathbf{R})\mathbf{p} - \frac{1}{2}\mathrm{Tr}\left[\mathbf{V}(\mathbf{R})\right]$$

H. D. Meyer and W. H. Miller, J. Chem. Phys., vol. 70, pp. 3214–3223, Apr. 1979.
G. Stock and M. Thoss, Phys. Rev. Lett., vol. 78, pp. 578–581, Jan. 1997.

Nonadiabatic semiclassical dynamics

$$e^{-\frac{i}{\hbar}\hat{H}t} \approx \iiiint \frac{d\mathbf{P}_0 d\mathbf{R}_0 d\mathbf{p}_0 d\mathbf{x}_0}{(2\pi\hbar)^{N+F}} C_t(\mathbf{z}_0) e^{\frac{i}{\hbar}S_t(\mathbf{z}_0)} \left|\mathbf{p}_t, \mathbf{x}_t\right\rangle \left|\mathbf{P}_t, \mathbf{R}_t\right\rangle \left\langle \mathbf{P}_0, \mathbf{R}_0\right| \left\langle \mathbf{p}_0, \mathbf{x}_0\right|$$

$$C_t(\mathbf{z}_0) = \sqrt{\left|\frac{1}{2}\left(\mathbb{M}_{\xi\xi}(t) + \Gamma_{N-el}^{-1}\mathbb{M}_{\pi\pi}(t)\Gamma_{N-el} - i\hbar\mathbb{M}_{\xi\pi}(t)\Gamma_{N-el} + \frac{i}{\hbar}\Gamma_{N-el}\mathbb{M}_{\pi\xi}(t)\right)\right|}$$

$$M(t) = \begin{pmatrix} M_{PP}(t) & M_{Pp}(t) & M_{PR}(t) & M_{Px}(t) \\ M_{pP}(t) & M_{pp}(t) & M_{pR}(t) & M_{px}(t) \\ M_{RP}(t) & M_{Rp}(t) & M_{RR}(t) & M_{Rx}(t) \\ M_{xP}(t) & M_{xp}(t) & M_{xR}(t) & M_{xx}(t) \end{pmatrix} = \begin{pmatrix} \mathbb{M}_{\pi\pi}(t) & \mathbb{M}_{\pi\xi}(t) \\ \mathbb{M}_{\xi\pi}(t) & \mathbb{M}_{\xi\xi}(t) \end{pmatrix}$$

$$S_t(\mathbf{z}_0) = \int_0^t d\tau \left[\mathbf{P}_\tau \dot{\mathbf{R}}_\tau + \mathbf{p}_\tau \dot{\mathbf{x}}_\tau - H(\mathbf{P}_\tau, \mathbf{R}_\tau, \mathbf{p}_\tau, \mathbf{x}_\tau)\right]$$

$$\left\langle \mathbf{y} \middle| \mathbf{p}, \mathbf{x}\right\rangle = \left(\frac{1}{\pi^N}\right)^{\frac{1}{4}} e^{-(\mathbf{y}-\mathbf{x})^T \mathbb{I}(\mathbf{y}-\mathbf{x}) + \frac{i}{\hbar}\mathbf{p}(\mathbf{y}-\mathbf{x})}$$

M. S. Church, T. J. H. Hele, G. S. Ezra, and N. Ananth, JCP, vol. 148, Dec. 2017
L. E. Cook, J. E. Runeson, J.O. Richardson and T. J. H. Hele; JCTC, 19, 18, 6109-6125
***D. Moscato**, M. Gandolfi and M. Ceotto; manuscript submitted*

The MInt algorithm

$$\Phi_{H,\delta t} = \Phi_{H_1,\delta t/2} \circ \Phi_{H_2,\delta t} \circ \Phi_{H_1,\delta t/2}$$

$$H_1(\mathbf{P}) = \sum_{J=1}^{F} \frac{P_J^2}{2m_J} \qquad H_2(\mathbf{R},\mathbf{p},\mathbf{x}) = V_0(\mathbf{R}) + \frac{1}{2}\mathbf{x}^T\mathbf{V}(\mathbf{R})\mathbf{x} + \frac{1}{2}\mathbf{p}^T\mathbf{V}(\mathbf{R})\mathbf{p} - \frac{1}{2}\mathrm{Tr}\left[\mathbf{V}(\mathbf{R})\right]$$

$$\begin{cases} \mathbf{x}(t+\delta t) = & \mathbb{C}(\mathbf{R})\mathbf{x}(t) + \mathbb{D}(\mathbf{R})\mathbf{p}(t) \\ \mathbf{p}(t+\delta t) = & \mathbb{C}(\mathbf{R})\mathbf{p}(t) - \mathbb{D}(\mathbf{R})\mathbf{x}(t) \\ P_k(t+\delta t) = & P_k(t) - \frac{\partial V_0}{\partial R_k}(\mathbf{R})\delta t - \frac{1}{2}\left[\mathbf{x}^T(t)\mathbb{E}_k(\mathbf{R})\mathbf{x}(t) + \mathbf{p}^T(t)\mathbb{E}_k(\mathbf{R})\mathbf{p}(t) - 2\mathbf{x}^T(t)\mathbb{F}_k(\mathbf{R})\mathbf{p}(t)\right] + \frac{1}{2}\mathrm{Tr}\left[\frac{\partial \mathbf{V}(\mathbf{R})}{\partial R_k}\right]\delta t \end{cases}$$

$$\frac{d}{dt}\left(\Phi_{H_1,t}M(0)\right) = \mathbf{J}\frac{\partial^2 H_1}{\partial \mathbf{z}^2}(\mathbf{z})M(t) \longrightarrow \dot{M}_{\mathbf{Rz}}(t) = m^{-1}M_{\mathbf{Pz}}(t)$$

$$\frac{d}{dt}\left(\Phi_{H_2,t}M(0)\right) = \mathbf{J}\frac{\partial^2 H_2}{\partial \mathbf{z}^2}(\mathbf{z})M(t) \longrightarrow \begin{cases} \dot{M}_{\mathbf{Pz}}(t) = & -\frac{\partial^2 H_2}{\partial \mathbf{R}^2}M_{\mathbf{Rz}}(t) - \frac{\partial^2 H_2}{\partial \mathbf{R}\partial \mathbf{p}}M_{\mathbf{pz}}(t) - \frac{\partial^2 H_2}{\partial \mathbf{R}\partial \mathbf{x}}M_{\mathbf{xz}}(t) \\ \dot{M}_{\mathbf{pz}}(t) = & -\mathbf{V}(\mathbf{R})M_{\mathbf{xz}}(t) - \left(\frac{\partial^2 H_2}{\partial \mathbf{R}\partial \mathbf{x}}\right)^T M_{\mathbf{Rz}}(t) \\ \dot{M}_{\mathbf{xz}}(t) = & \mathbf{V}(\mathbf{R})M_{\mathbf{pz}}(t) + \left(\frac{\partial^2 H_2}{\partial \mathbf{R}\partial \mathbf{p}}\right)^T M_{\mathbf{Rz}}(t) \end{cases}$$

M. S. Church, T. J. H. Hele, G. S. Ezra, and N. Ananth, JCP, vol. 148, Dec. 2017
L. E. Cook, J. E. Runeson, J.O. Richardson and T. J. H. Hele; JCTC, 19, 18, 6109-6125
***D. Moscato**, M. Gandolfi and M. Ceotto; paper in preparation*

Nonadiabatic semiclassical spectroscopy

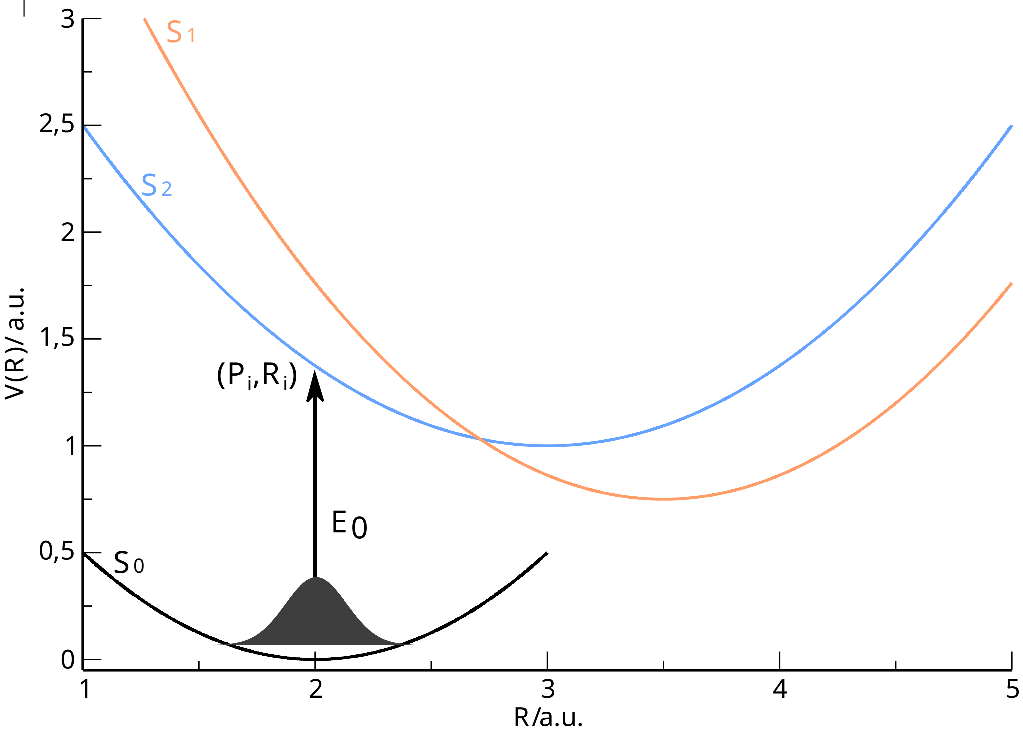
$$I(E) \approx \iiiint \frac{d\mathbf{P}_0 d\mathbf{R}_0 d\mathbf{p}_0 d\mathbf{x}_0}{(2\pi\hbar)^{N+F}} \frac{1}{2\mathrm{T}} \left| \int_0^{\mathrm{T}} dt\; e^{\frac{i}{\hbar}[S_t(\mathbf{z}_0)+\phi_t(\mathbf{z}_0)+(E+E_0)t]} \langle \psi_0 | \mathbf{z}_t \rangle \right|^2$$

$$|\psi_0\rangle = \sum_{l=1}^{N} \sqrt{\rho_{ll}(p_{0,l}, x_{0,l})}\, |0_1, \ldots, 1_l, \ldots, 0_N\rangle |\mathbf{P}_{eq}^{(0)}, \mathbf{R}_{eq}^{(0)}\rangle$$

Focused initial conditions

$$\sum_{l}^{N} \rho_{ll}(0) = \sum_{l}^{N} \frac{1}{2}(x_l^2(0) + p_l^2(0) - 1) = 1$$

$$\begin{cases} p_{0,l} = \sqrt{2\rho_{ll}(0)+1}\,\sin(\theta_{0,l}) \\ x_{0,l} = \sqrt{2\rho_{ll}(0)+1}\,\cos(\theta_{0,l}) \end{cases}$$

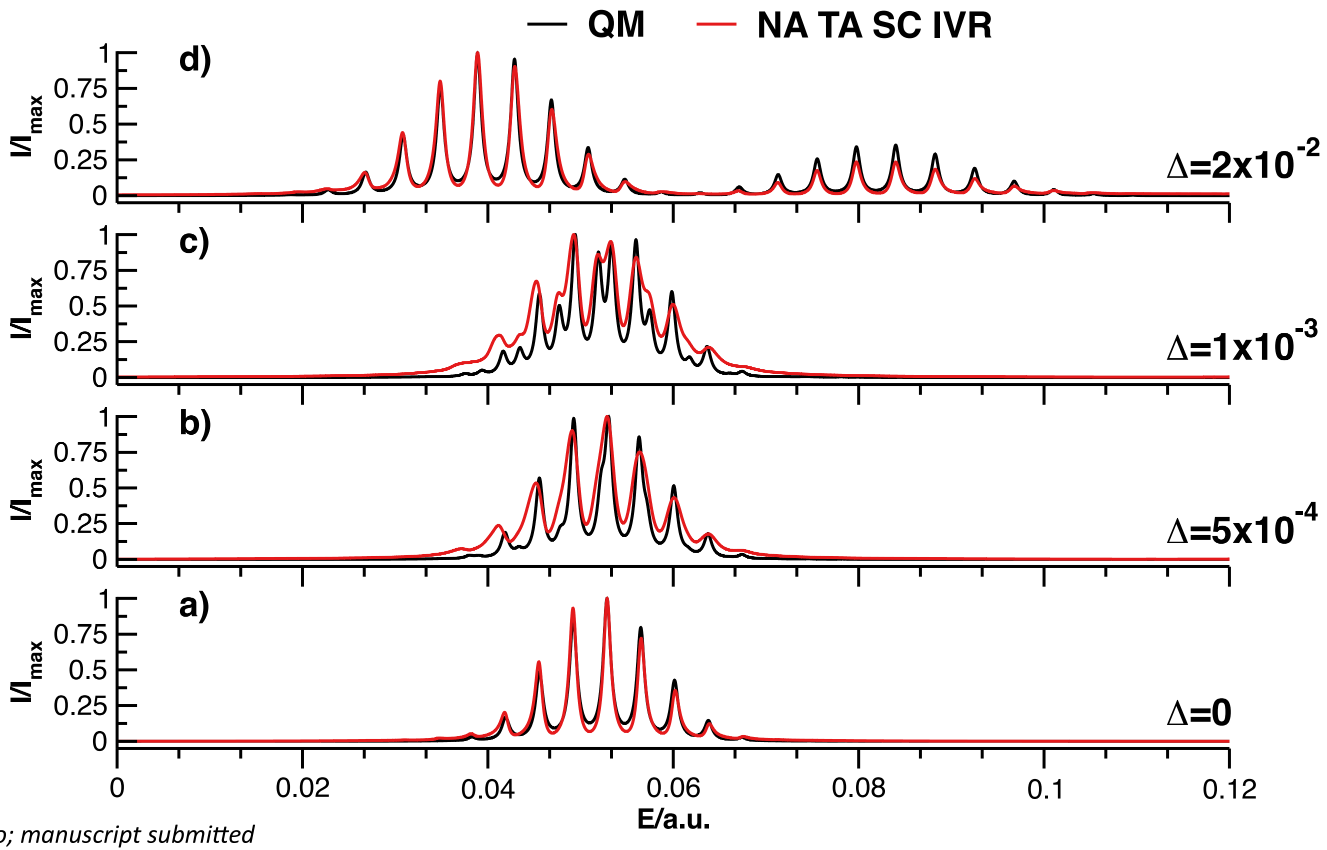


S. Bonella and D. F. Coker, JCP, vol. 118, p. 4370–4385, Mar. 2003.
***D. Moscato**, M. Gandolfi and M. Ceotto; manuscript submitted*

Nonadiabatic semiclassical spectroscopy

$$\mathbf{V}(R) = \begin{vmatrix} \frac{1}{2}\omega_1^2 (R - R_{eq,1})^2 + \varepsilon_1 & \Delta \\ \Delta & \frac{1}{2}\omega_2^2 (R - R_{eq,2})^2 + \varepsilon_2 \end{vmatrix}$$

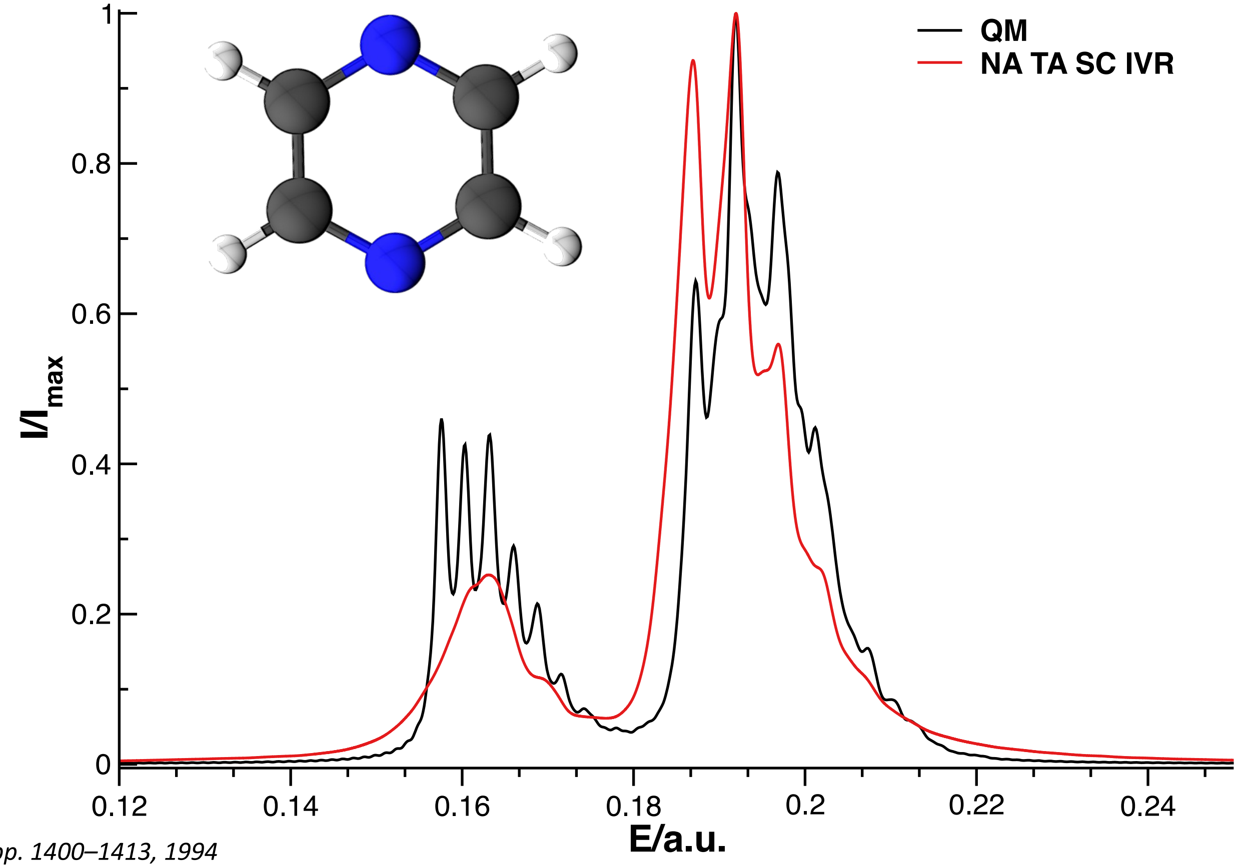
$$\omega_1 = 0.0045563 \text{ a.u.}, \ R_{eq,1} = 0 \text{ a.u.}, \ \varepsilon_1 = 0.36451 \text{ a.u.}$$
$$\omega_2 = 0.0036450 \text{ a.u.}, \ R_{eq,2} = 10.0 \text{ a.u.}, \ \varepsilon_1 = \varepsilon_2$$



***D. Moscato**, M. Gandolfi and M. Ceotto; manuscript submitted*

Nonadiabatic semiclassical spectroscopy

$$\hat{H} = \sum_{J \in G_1, \nu_{10a}} \left[\hat{T}_J + \frac{\omega_J}{2} R_J^2 \right] \mathbb{1} + \begin{vmatrix} E_1 + \sum_{J \in G_1} \kappa_J^{(1)} R_J & \lambda R_{\nu_{10a}} \\ \lambda R_{\nu_{10a}} & E_2 + \sum_{J \in G_1} \kappa_J^{(2)} R_J \end{vmatrix}$$



C. Woywod, W. Domcke, A. L. Sobolewski, and H.J. Werner, JCP, vol. 100, no. 2, pp. 1400–1413, 1994
A. R. Walton, D. E. Manolopolous, Mol. Phys., 87(4), 961-978., 1995
M. Thoss, W. H. Miller, and G. Stock, JCP, vol. 112, p. 10282–10292, June 2000.
***D. Moscato**, M. Gandolfi and M. Ceotto; manuscript submitted*

Conclusions and outlooks

- SC IVR is a practical method to add NQEs to classical MD
- DC SCIVR allows for calculation of vibrational spectra of large molecular system with high accuracy as long as the potential is accurate
- NA TA SCIVR still has room for improvement

What’s next ?

- NA TA SCIVR in MMST adiabatic representation
- Application to atomistic systems
- Move beyond the mean-field NAMD ?

Conclusions

Dr. G. Mandelli

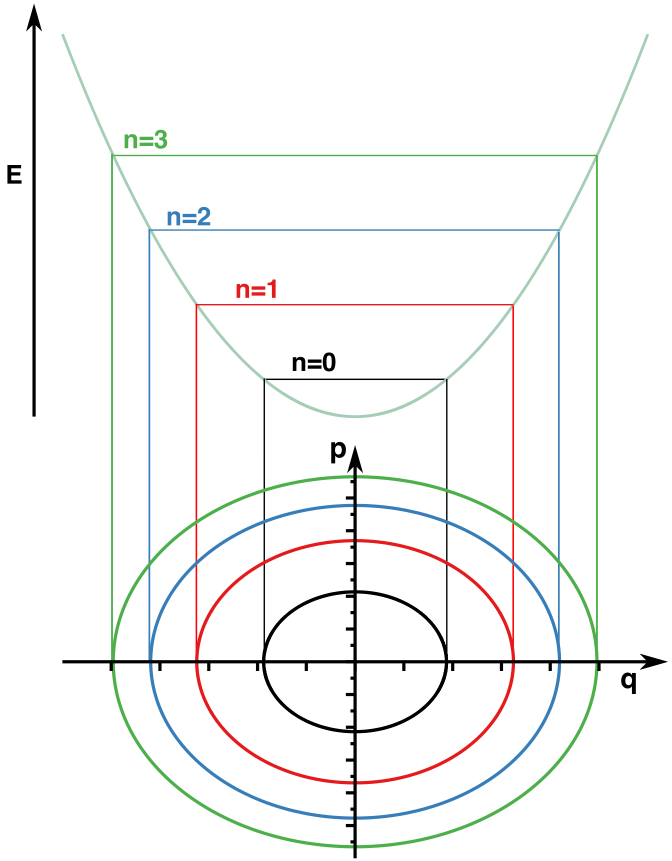
Dr. M. Gandolfi

Acknowledgements:

CINECA

Iscra_C _SemDynHP

Supplemental information #1: Quasiclassical Trajectory



$$\begin{cases} p_J(0) & = \sqrt{\hbar\omega_J(2n_J+1)}\,\sin(\theta_J) \\ q_J(0) & = q_{J,eq} + \sqrt{\frac{\hbar}{\omega_J}(2n_J+1)}\,\cos(\theta_J) \end{cases}$$

$$\downarrow$$

$$I_J(\omega) = \frac{1}{2\mathrm{T}}\left|\int_0^{\mathrm{T}} dt\ e^{i\omega t} p_J(t)\right|^2$$

N. De Leon and E. J. Heller, JCP, 78, 4005–4017, Mar. 1983.
J. M. Bowman, A. Kuppermann, and G. C. Schatz, JCPL, 19, 1, 21–25, 1973.
J. M. Bowman, B. Gazdy, and Q. Sun, JCP, 91,2859–2862, Sept. 1989.

Supplemental information #2: The HHKK propagator

$$e^{-\frac{i}{\hbar}\hat{H}t} \approx \iint \frac{d\mathbf{p}_0 d\mathbf{q}_0}{(2\pi\hbar)^F} C_t(\mathbf{p}_0,\mathbf{q}_0)\big|\mathbf{p}_t,\mathbf{q}_t\big\rangle\big\langle\mathbf{p}_0,\mathbf{q}_0\big|e^{\frac{i}{\hbar}S_t(\mathbf{p}_0,\mathbf{q}_0)}$$

$$\langle\mathbf{x}|\mathbf{p},\mathbf{q}\rangle = \left(\frac{\det\Gamma}{\pi^F}\right)e^{-(\mathbf{x}-\mathbf{q})^T\Gamma(\mathbf{x}-\mathbf{q})+\frac{i}{\hbar}\mathbf{p}(\mathbf{x}-\mathbf{q})} \longrightarrow \Gamma_{ij} = \omega_i\delta_{ij}$$

$$C_t(\mathbf{p}_0,\mathbf{q}_0) = \sqrt{\left|\frac{1}{2}\left(\frac{\partial\mathbf{q}_t}{\partial\mathbf{q}_0} + \Gamma^{-1}\frac{\partial\mathbf{p}_t}{\partial\mathbf{p}_0}\Gamma - i\hbar\frac{\partial\mathbf{q}_t}{\partial\mathbf{p}_0}\Gamma + \frac{i\Gamma^{-1}}{\hbar}\frac{\partial\mathbf{p}_t}{\partial\mathbf{q}_0}\right)\right|}$$

$$\Big\downarrow$$

$$\dot{M}(t) = \mathbf{J}\frac{\partial^2 H}{\partial\mathbf{z}^2}(\mathbf{z}(t))M(t)$$

J. H. Van Vleck, PNAS, vol. 14, p. 178–188, Feb. 1928.
W. H. Miller, JCP, vol. 53, p. 3578–3587, Nov. 1970.
E. J. Heller, JCP, vol. 75, p. 2923–2931, Sept. 1981.
M. F. Herman and E. Kluk, Chemical Physics, vol. 91, p. 27–34, Nov. 1984.
K. G. Kay, JCP, vol. 101, p. 2250–2260, Aug. 1994.

Supplemental information #3: Polarizable Embedding QM/MM

$$E(\mathbf{q},\mathbf{D},\boldsymbol{\mu}_d,\boldsymbol{\mu}_p) = E_{QM}(\mathbf{q},\mathbf{D}) + E_{env}(\mathbf{q},\mathbf{D},\boldsymbol{\mu}_d,\boldsymbol{\mu}_p)$$

$$E_{env}(\mathbf{q},\mathbf{D},\boldsymbol{\mu}_d,\boldsymbol{\mu}_p) = E_{MM}(\mathbf{q}) + E_{pol}(\mathbf{q},\mathbf{D},\boldsymbol{\mu}_d,\boldsymbol{\mu}_p) + E_{QM/MM}(\mathbf{q},\mathbf{D})$$

$$E_{pol}(\mathbf{q},\mathbf{D},\boldsymbol{\mu}_d,\boldsymbol{\mu}_p) = \frac{1}{2}\boldsymbol{\mu}_d^T\mathcal{T}\boldsymbol{\mu}_p - \frac{1}{2}\left(\boldsymbol{\mu}_p^T\mathbf{E}_d + \boldsymbol{\mu}_d^T\mathbf{E}_p\right) - \frac{1}{2}\left(\boldsymbol{\mu}_p + \boldsymbol{\mu}_d\right)^T\mathbf{E}_{QM}(\mathbf{Q},\mathbf{D})$$

$$\downarrow$$

$$\tilde{\mathbf{F}} = \frac{\partial E}{\partial \mathbf{D}}(\mathbf{q},\mathbf{D},\boldsymbol{\mu}_d,\boldsymbol{\mu}_p) = \mathbf{F}^0 + \mathbf{F}_{env}$$

D. Loco, E. Polack, S. Caprasecca, L. Lagardère, F. Lipparini, J.P Piquemal, B. Mennucci; JCTC 2016, 12, 8, 3654–3661
D. Loco, L. Lagardère, S. Caprasecca, F. Lipparini, B. Mennucci, J.P. Piquemal; JCTC 2017, 13, 4025-4033

Supplemental information #4: The MInt algorithm

$$\Phi_{H,\delta t} = \Phi_{H_1,\delta t/2} \circ \Phi_{H_2,\delta t} \circ \Phi_{H_1,\delta t/2}$$

$$H_1(\mathbf{P}) = \sum_{J=1}^{F} \frac{P_J^2}{2M_J}$$

$$H_2(\mathbf{R},\mathbf{p},\mathbf{x}) = V_0(\mathbf{R}) + \frac{1}{2}\mathbf{x}^T\mathbf{V}(\mathbf{R})\mathbf{x} + \frac{1}{2}\mathbf{p}^T\mathbf{V}(\mathbf{R})\mathbf{p} - \frac{1}{2}\mathrm{Tr}\left[\mathbf{V}(\mathbf{R})\right]$$

M. S. Church, T. J. H. Hele, G. S. Ezra, and N. Ananth, JCP, vol. 148, Dec. 2017
L. E. Cook, J. E. Runeson, J.O. Richardson and T. J. H. Hele; JCTC, 19, 18, 6109-6125
***D. Moscato**, M. Gandolfi and M. Ceotto; paper in preparation*

Supplemental information #4: The MInt algorithm

$$\Phi_{H,\delta t} = \Phi_{H_1,\delta t/2} \circ \Phi_{H_2,\delta t} \circ \Phi_{H_1,\delta t/2}$$

$$\mathbf{R}\left(t + \frac{\delta t}{2}\right) = \mathbf{R}(t) + \mathbf{M}^{-1}\mathbf{P}(t)\frac{\delta t}{2}$$

$$\frac{d}{dt}\left(\Phi_{H_1,t} M(0)\right) = \mathbf{J}\frac{\partial^2 H_1}{\partial \mathbf{z}^2}(\mathbf{z})M(t) \longrightarrow \dot{M}_{\mathbf{Rz}}(t) = m^{-1}M_{\mathbf{Pz}}(t)$$

M. S. Church, T. J. H. Hele, G. S. Ezra, and N. Ananth, JCP, vol. 148, Dec. 2017
L. E. Cook, J. E. Runeson, J.O. Richardson and T. J. H. Hele; JCTC, 19, 18, 6109-6125
***D. Moscato**, M. Gandolfi and M. Ceotto; paper in preparation*

Supplemental information #4: The MInt algorithm

$$\begin{cases} \mathbf{x}(t+\delta t) = & \mathbb{C}(\mathbf{R})\mathbf{x}(t) + \mathbb{D}(\mathbf{R})\mathbf{p}(t) \\ \mathbf{p}(t+\delta t) = & \mathbb{C}(\mathbf{R})\mathbf{p}(t) - \mathbb{D}(\mathbf{R})\mathbf{x}(t) \\ P_K(t+\delta t) = & P_K(t) - \frac{\partial V_0}{\partial R_K}(\mathbf{R})\delta t - \frac{1}{2}\left[\mathbf{x}^T(t)\mathbb{E}_K(\mathbf{R})\mathbf{x}(t) + \mathbf{p}^T(t)\mathbb{E}_K(\mathbf{R})\mathbf{p}(t) - 2\mathbf{x}^T(t)\mathbb{F}_K(\mathbf{R})\mathbf{p}(t)\right] + \frac{1}{2}\mathrm{Tr}\left[\frac{\partial \mathbb{V}(\mathbf{R})}{\partial R_K}\right]\delta t \end{cases}$$

$$\Lambda(\mathbf{R}) = \mathbb{S}^T(\mathbf{R})\mathbf{V}(\mathbf{R})\mathbb{S}(\mathbf{R})$$

$$\mathbb{C}(\mathbf{R}) = \mathbb{S}\cos\left(\delta t \Lambda(\mathbf{R})\right)\mathbb{S}^T(\mathbf{R}) = \cos\left(\delta t \mathbb{V}(\mathbf{R})\right)$$

$$\mathbb{D}(\mathbf{R}) = \mathbb{S}\sin\left(\delta t \Lambda(\mathbf{R})\right)\mathbb{S}^T(\mathbf{R}) = \sin\left(\delta t \mathbb{V}(\mathbf{R})\right)$$

$$(\Gamma_k)_{lm} = \begin{cases} (\mathbb{W}_k)_{lm}\,\delta t & m = n \\ \frac{(\mathbb{W}_k)_{lm}}{\lambda_{lm}}\sin\left(\lambda_{lm}\delta t\right) & m \neq n \end{cases}$$

$$(\Omega_k)_{lm} = \begin{cases} 0 & m = n \\ \frac{(\mathbb{W}_k)_{lm}}{\lambda_{lm}}\left[1 - \cos\left(\lambda_{lm}\delta t\right)\right] & m \neq n \end{cases}$$

$$\mathbb{E}_k(\mathbf{R}) = \mathbb{S}(\mathbf{R})\Gamma_k(\mathbf{R})\mathbb{S}^T(\mathbf{R})$$

$$\mathbb{F}_k(\mathbf{R}) = \mathbb{S}(\mathbf{R})\Omega_k(\mathbf{R})\mathbb{S}^T(\mathbf{R})$$

M. S. Church, T. J. H. Hele, G. S. Ezra, and N. Ananth, JCP, vol. 148, Dec. 2017
L. E. Cook, J. E. Runeson, J.O. Richardson and T. J. H. Hele; JCTC, 19, 18, 6109-6125

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$$\Phi_{H,\delta t} = \Phi_{H_1,\delta t/2} \circ \Phi_{H_2,\delta t} \circ \Phi_{H_1,\delta t/2}$$

$$\begin{cases} \mathbf{x}(t+\delta t) = & \mathbb{C}(\mathbf{R})\mathbf{x}(t) + \mathbb{D}(\mathbf{R})\mathbf{p}(t) \\ \mathbf{p}(t+\delta t) = & \mathbb{C}(\mathbf{R})\mathbf{p}(t) - \mathbb{D}(\mathbf{R})\mathbf{x}(t) \\ P_k(t+\delta t) = & P_k(t) - \frac{\partial V_0}{\partial R_k}(\mathbf{R})\delta t - \frac{1}{2}\left[\mathbf{x}^T(t)\mathbb{E}_k(\mathbf{R})\mathbf{x}(t) + \mathbf{p}^T(t)\mathbb{E}_k(\mathbf{R})\mathbf{p}(t) - 2\mathbf{x}^T(t)\mathbb{F}_k(\mathbf{R})\mathbf{p}(t)\right] + \frac{1}{2}\mathrm{Tr}\left[\frac{\partial \mathbf{V}(\mathbf{R})}{\partial R_k}\right]\delta t \end{cases}$$

$$\frac{d}{dt}\left(\Phi_{H_2,t}M(0)\right) = \mathbf{J}\frac{\partial^2 H_2}{\partial \mathbf{z}^2}(\mathbf{z})M(t) \longrightarrow \begin{cases} \dot{M}_{\mathbf{Pz}}(t) = & -\frac{\partial^2 H_2}{\partial \mathbf{R}^2}M_{\mathbf{Rz}}(t) - \frac{\partial^2 H_2}{\partial \mathbf{R}\partial \mathbf{p}}M_{\mathbf{pz}}(t) - \frac{\partial^2 H_2}{\partial \mathbf{R}\partial \mathbf{x}}M_{\mathbf{xz}}(t) \\ \dot{M}_{\mathbf{pz}}(t) = & -\mathbf{V}(\mathbf{R})M_{\mathbf{xz}}(t) - \left(\frac{\partial^2 H_2}{\partial \mathbf{R}\partial \mathbf{x}}\right)^T M_{\mathbf{Rz}}(t) \\ \dot{M}_{\mathbf{xz}}(t) = & \mathbf{V}(\mathbf{R})M_{\mathbf{pz}}(t) + \left(\frac{\partial^2 H_2}{\partial \mathbf{R}\partial \mathbf{p}}\right)^T M_{\mathbf{Rz}}(t) \end{cases}$$

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L. E. Cook, J. E. Runeson, J.O. Richardson and T. J. H. Hele; JCTC, 19, 18, 6109-6125
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